THREE-DIMENSIONAL SLOPE STABILITY COMPARATIVE CASE STUDY

A Limit-Equilibrium Study Using Limestone Topography and Global Geomechanical Variations

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Data Engine Baseline: USGS Scoops3D (v1.3.01) | Spatial Reference Control: EPSG:32632 (WGS 84 / UTM Zone 32N)

1. Detailed Methodology: From Photogrammetry to 3D Limit-Equilibrium Solver

In this project I built a multi-stage workflow that connects drone photogrammetry data to a three-dimensional limit-equilibrium slope stability model. The main strength of the approach is that the topography stays fixed as a control variable, so any change in the results comes only from the strength, groundwater and seismic inputs of the different lithologies and climates.

1.1 UAV Photogrammetry & Mesh Preprocessing (Pix4Dmapper)

Dataset Source: I did not capture the drone imagery myself. I used the public “Quarry” example dataset that Pix4D provides on its support site, which Pix4D offers for personal or professional training (courtesy of Pix4D / pix4d.com). Pix4D describes the site only as a quarry in Switzerland. The coordinates of my final DEM (EPSG:32632) fall at about 46.65° N, 6.54° E, which is the Eclépens quarry area in the Canton of Vaud, so I refer to it as the Eclépens Limestone Quarry (Carrière d’Eclépens) in this study. This surface is the geometric base for every test scenario.

Image Acquisition (from the Pix4D dataset): The dataset was flown with a senseFly eBee Classic fixed-wing UAV and a senseFly S.O.D.A. camera, in one flight with a double-grid flight plan. It has 347 images of 5472 × 3648 pixels, geolocated in WGS84, with an average ground sampling distance (GSD) of 5.40 cm over an area of 0.5776 km² (57.76 ha).

Some Images of the Dataset:



Figure 1: Image 1



Figure 2: Image 2



Figure 3: Image 3

<u>Ground Control and Processing</u>: The dataset includes 9 3D Ground Control Points (GCPs) in the Swiss national coordinate system (CH1903 / LV03), which fix the spatial accuracy of the model. I processed the images in Pix4Dmapper, and the outputs stayed in CH1903 / LV03 until I reprojected them in the next step.

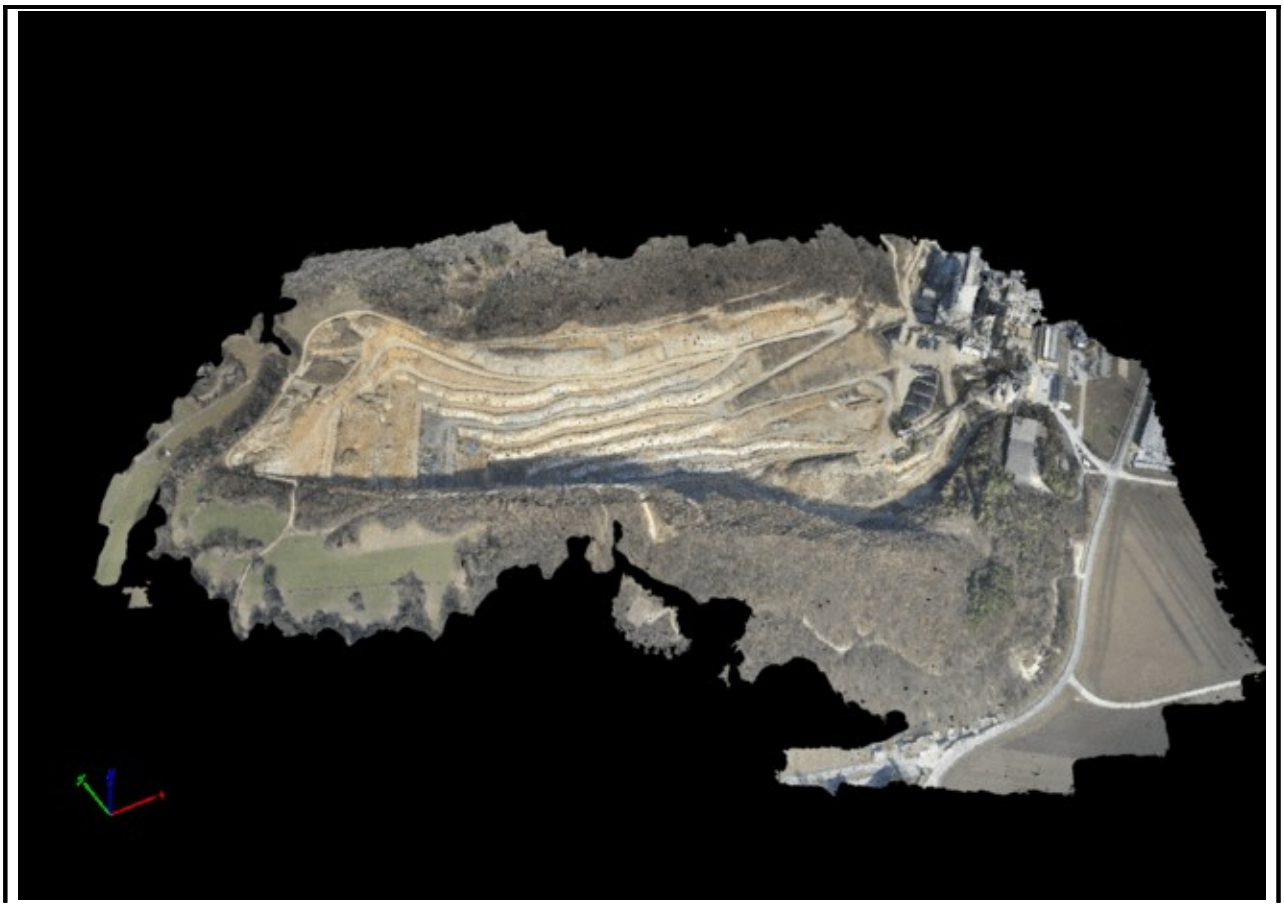


Figure 4: Automated 3D point cloud mesh generation of the active Eclépens Limestone Quarry via Pix4Dmapper drone photogrammetry. Dataset courtesy of Pix4D / pix4d.com.

1.2 Spatial Coordinate Reprojection & Raster Downsampling (QGIS)

***Projection Translation*:** The local survey coordinates were not suitable for a regional slope solver, which needs a continuous metric grid. So I imported the raw digital elevation raster into QGIS and reprojected it to EPSG:32632 (WGS 84 / UTM Zone 32N).

***Grid Size Optimization*:** The original 1-meter Digital Surface Model (DSM) was very dense and was computationally heavy for my system, so I downsampled it to a uniform 10-meter cell size and saved it as an ASCII grid (quarry.asc), using the argument *(-tr 10.0 10.0)* in the QGIS GDAL Translate tool.

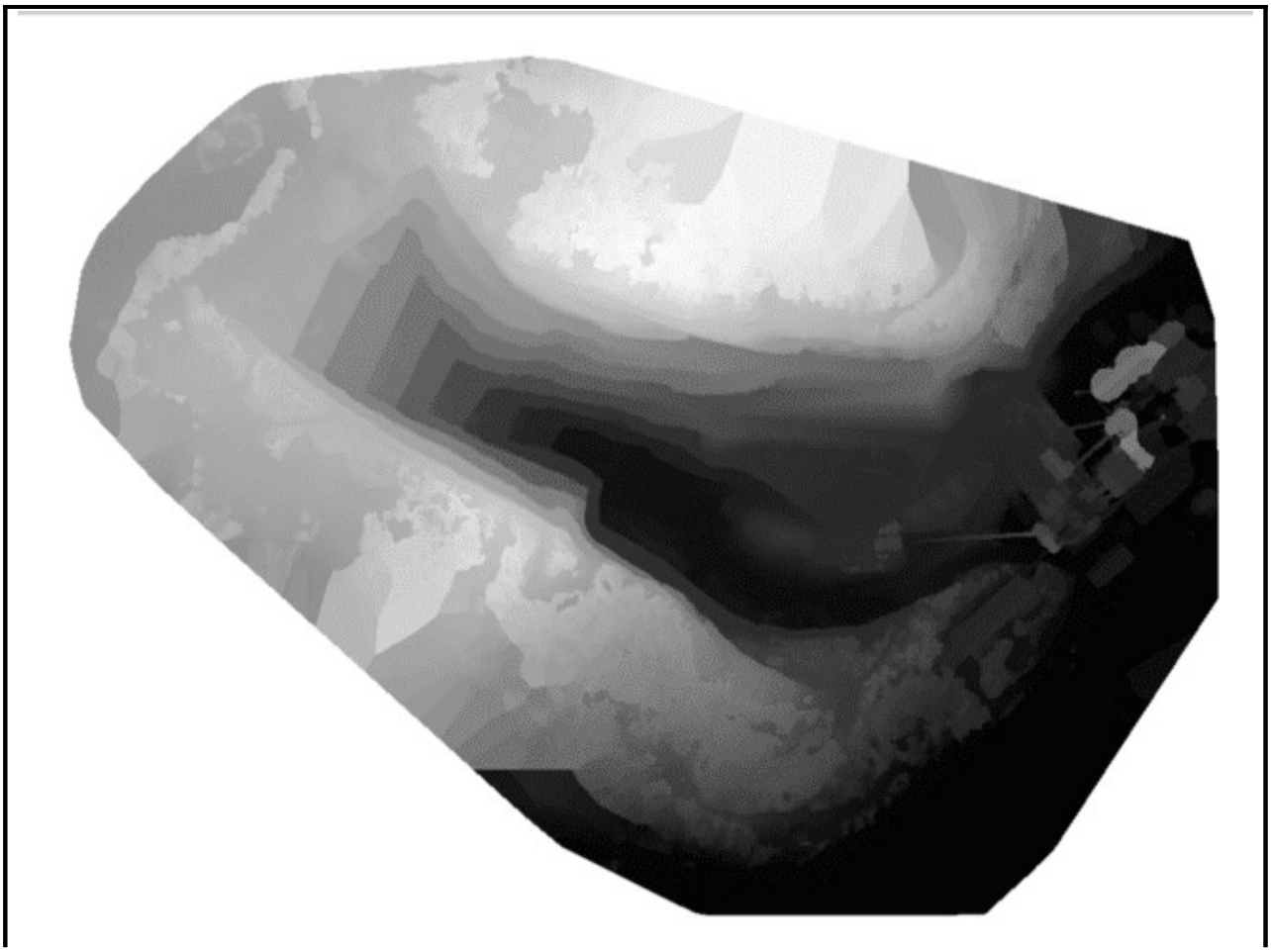


Figure 5: Downsampled 10-meter cell size Digital Surface Model (DSM) ASCII grid format reprojected to EPSG:32632.

***Grid Discretization*:** The coarser cell size smooths out drone surface texture, vegetation noise and dust that would otherwise disturb the column equations, and it gives much more stable slope stability results. The final computational domain is a matrix of ismax = 131 columns by jsmax = 98 rows, with **9,299** active non-null cells used in the calculation.

1.3 3D Limit-Equilibrium Mathematical Execution (Scoops3D)

***The Command Engine Processing Pipeline*:** I ran the 10m ASCII grid directly in the *USGS Scoops3D* command-line executable (Scoops3D_1.3.01win64.exe), which avoided formatting conflicts with the user interface text script.

***The Mathematical Framework Solver*:** Rotational stability was calculated with Bishop's Simplified 3D Method of Columns. The program divides each potential spherical slide volume into many square vertical columns and works out the weight, normal forces and shear resistance along the 3D base of every column [Reid et al., 2015].

Latticed Search Optimization Parameters:

• Search Method (srch = box): The search lattice covered the whole DEM (nodes 1 to 131 in x and 1 to 98 in y), with a radius increment of dr = 5 m.

• Lattice Spacing Alignment: The search spacing was matched to the terrain resolution: nsrchres = 1 (one search node per 10 m DEM cell) and a vertical spacing of zsrchres = 10.0m, between zsmin = 450 m and zsmax = 700 m.

• Size Constraint Threshold Filters: To leave out minor bench raveling and trivial rockfall, I filtered the potential failures strictly by volume (vacriterion = Volume), with a minimum (vmin) of 50,000 m^3 and a maximum (vmax) of 100,000 m^3.

• Volume Tolerance and Column Limit: The volume tolerance was tol = 5,000 m^3 in all runs. The minimum number of active columns was limcol = 100. Scoops3D lists every trial surface with fewer active columns in the ncolerr output files, which are discussed in the note under the table in Section 4.

1.4 Post-Processing & Heatmap Symbology Styling

I loaded the computed Factor of Safety grids (*_fos3d_out.asc) back into QGIS and styled them to get high-contrast hazard maps:

Coordinate Layer Lock: I manually set the layer coordinates to EPSG:32632 so the calculated safety grid lines up over the base quarry topography.

Symbology Threshold Settings: I changed the render type to Singleband pseudocolor with an inverted, high-contrast RdYlGn (Red-Yellow-Green) palette.

Data Range Normalization: The empty background and air cells have a default value of 100 from the program, and this flattens the color scale. To avoid that, I fixed the maximum range at 4.0 before clicking Classify. This puts the whole color ramp on the active slope faces: critical failures below 1.1 show as bright red, marginal areas as yellow, and fully safe zones as solid green.

2. Geotechnical Criteria Design Coefficients

The table below gives the material properties, water pressure ratios and seismic coefficients I used to separate each country scenario. These are representative values for each type of environment, these values were taken from *Google AI Mode* and are not validated against real values, so the results show how each type of environment behaves and not the actual condition of the named mines.

Country / Site	Cohesion (c')	Friction (φ')	Unit Wt (γ)	Water (Ru)	Seismic (kh)	Environmental Profile Summary
Switzerland (Baseline)	140.0 kPa	38.0°	25.5 kN/m^3	0.20	0.00	Competent limestone mass, affected by seasonal Alpine spring snowmelt.
Zambia (Nchanga)	35.0 kPa	27.0°	23.0 kN/m^3	0.35	0.00	Highly weathered, jointed sub-tropical shales with low-cohesion

						saprolitic zones.
India (Noamundi)	25.0 kPa	24.0°	21.5 kN/m³	0.40	0.00	Highly degraded lateritic clay overburden exposed to severe monsoons.
Kazakhstan (Bozshakol)	25.0 kPa	34.0°	24.5 kN/m³	0.10	0.00	Sheared sedimentary rock mass under low groundwater in a semi-arid climate.
Mongolia (Oyu Tolgoi)	180.0 kPa	42.0°	26.5 kN/m³	0.00	0.12	High-strength dry rock mass under tectonic earthquake loading.

3. Geotechnical & Environment-Driven Heatmap Behavior Analysis

3.1 Switzerland: Eclépens Limestone Quarry Baseline (The Control Run)

Heatmap Behavioral Pattern: The outer rim and pit margins are stable, with deep green zones running continuously around them. Orange and red zones cover the central pit area, with the strongest red along the eastern edge, along a diagonal band through the pit and at one patch near the lower-left edge.

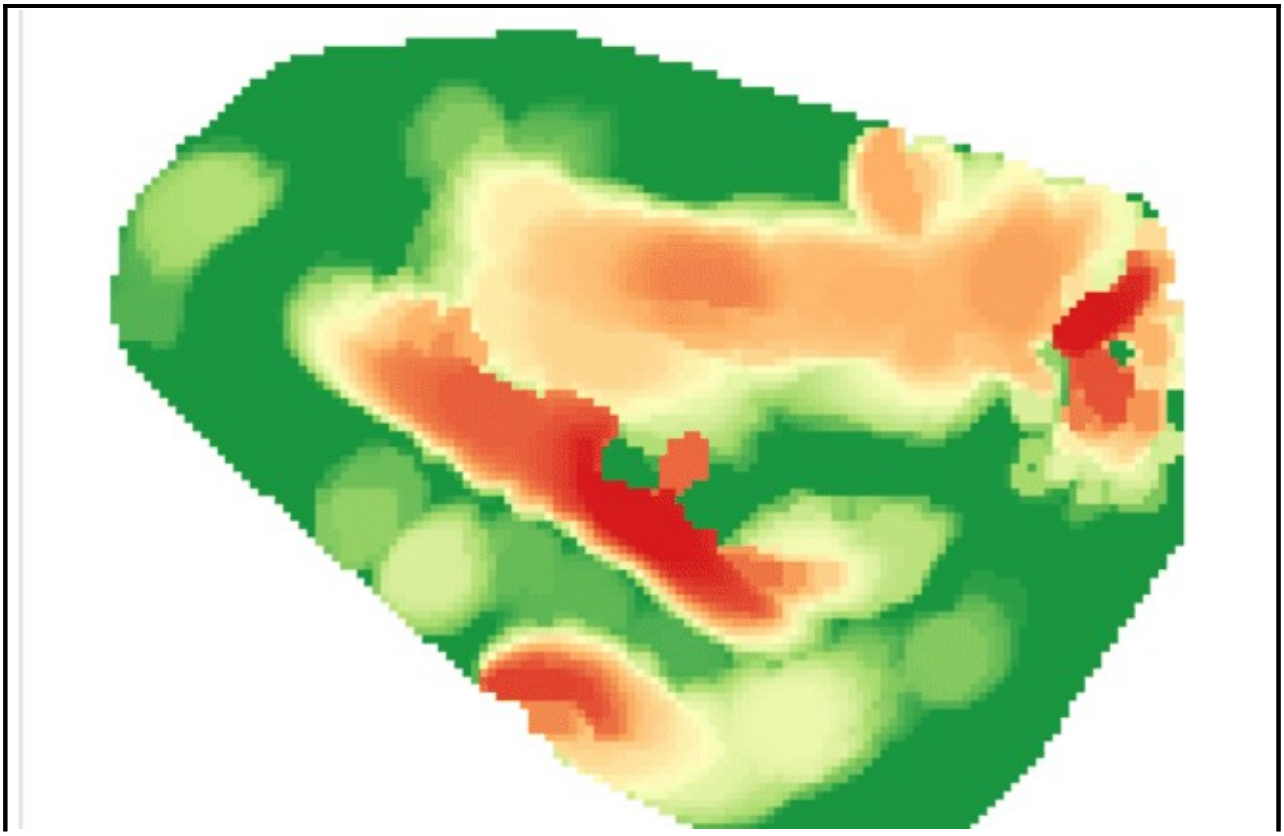


Figure 6: Factor of Safety (FoS) 3D baseline control run heatmap showing stable green outer margins, with orange and red zones in the steepest parts of the central pit, under the limestone parameters.

***Geomechanical Discussion*:** The limestone has the highest strength of the five scenarios in this study (c' = 140.0 kPa, φ' = 38.0°), so most of the quarry stays stable even with its steep, benched geometry, and the Alpine meltwater coefficient of Ru = 0.20 adds only minor pore-water pressure. The baseline is not free of failures, however. The minimum Bishop Factor of Safety is 0.7346, which is below 1, and the orange and red zones in the central pit show that the steepest wall sections are unstable even in this strong rock. The outer margins are stable, but the baseline is the least severe result in the study and not a fully stable one.

3.2 Zambia: Nchanga Open Pit (Weathered Tropical Setup)

***Heatmap Behavioral Pattern*:** The stable green field breaks down sharply. Wide, overlapping orange and red zones spread across the central floor and the lower multi-bench slopes of the highwall.

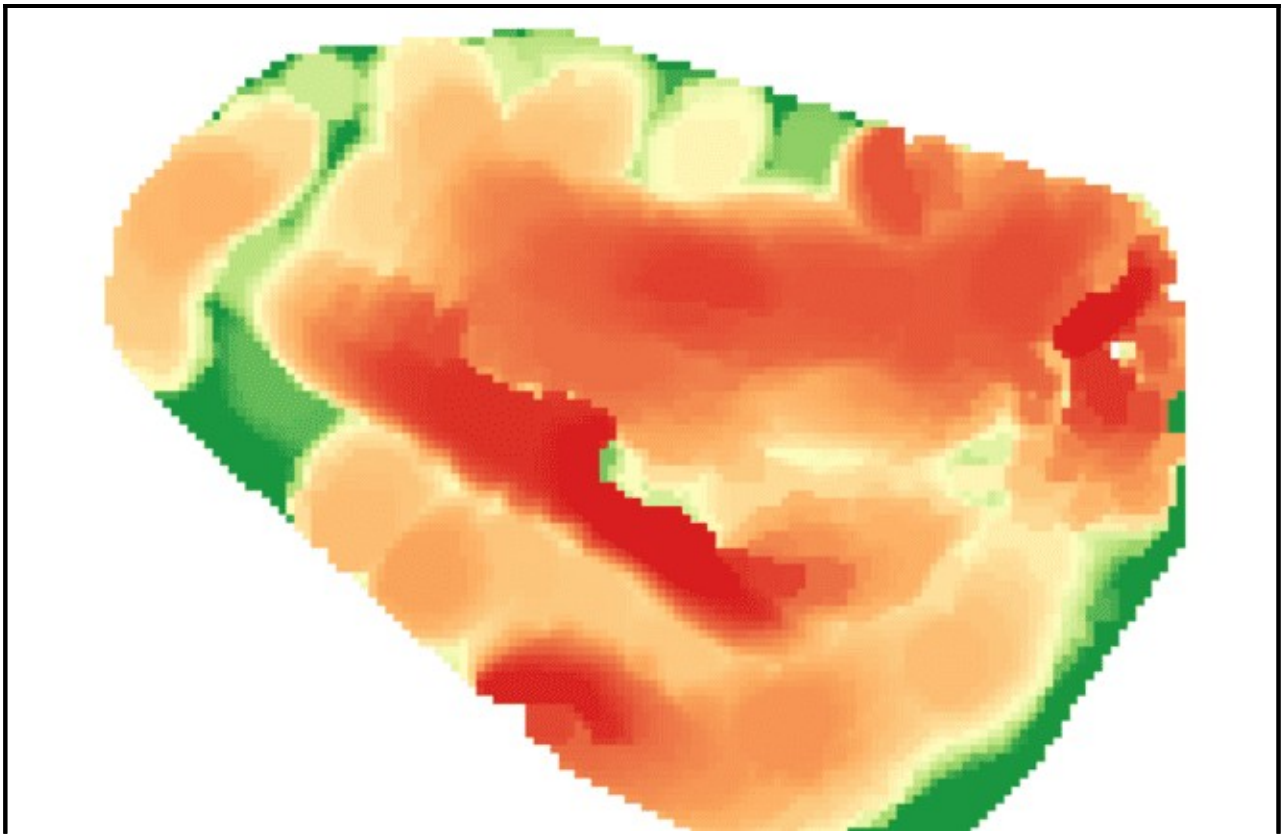


Figure 7: Factor of Safety heatmap for the simulated Zambia Nchanga setup, indicating a widespread stability drop across the lower highwall benches.

Geomechanical Discussion: This big change shows how rock strength drops in a humid, sub-tropical climate. The low cohesion (c' = 35.0 kPa) and low friction angle (φ' = 27.0°) describe a highly weathered, saprolitic rock mass that is too weak to hold steep angles. With the high pore pressure (Ru = 0.35), the trapped water pushes the joints and fractures apart and reduces the effective shear resistance. The safety factor drops sharply and large parts of the central quarry slope move toward failure.

How the Real Mine Layout Copes (From the Photograph)



Figure 8: Photograph of the Nchanga Open Pit (Zambia), showing wide benches and a ramp in layered rock, with vegetation on some benches and water collected in a small pond on the pit floor.

The photograph of the Nchanga open pit shows the features below, which help it cope with weak, wet rock. These are observations from the photograph only, and the angles quoted come from the screening check in Section 6.

- **Wide Benches Break the Wall Into Short Faces:** The wall is made of wide benches and a ramp, not one long steep face. In the screening check in Section 6, material with these strength values (Ru = 0.35) can only be cut at about 36° for a single 15 m bench and about 16° for a 150 m wall to keep a FoS of at least 1.3, so wide benches are what keep the overall angle low.

- **Water Collects on the Pit Floor:** A small pond has collected on the floor of the pit, which shows that water ends up at the bottom and not only inside the walls. Keeping water out of the walls is what lowers Ru: in the screening check, lowering Ru from 0.35 to 0.10 raises the limits to about 49° for a bench and 23° for the wall. The photograph does not show whether pumps or dewatering wells are used.

3.3 India: Noamundi Mine (Extreme Monsoon-Saturated Setup)

***Heatmap Behavioral Pattern*:** This is the most severe failure case in the whole project. Orange and red zones cover almost the entire inner pit and run up the main highwall faces. The strongest red follows a diagonal band through the pit and the eastern edge, and only small green areas remain near the outer edge.

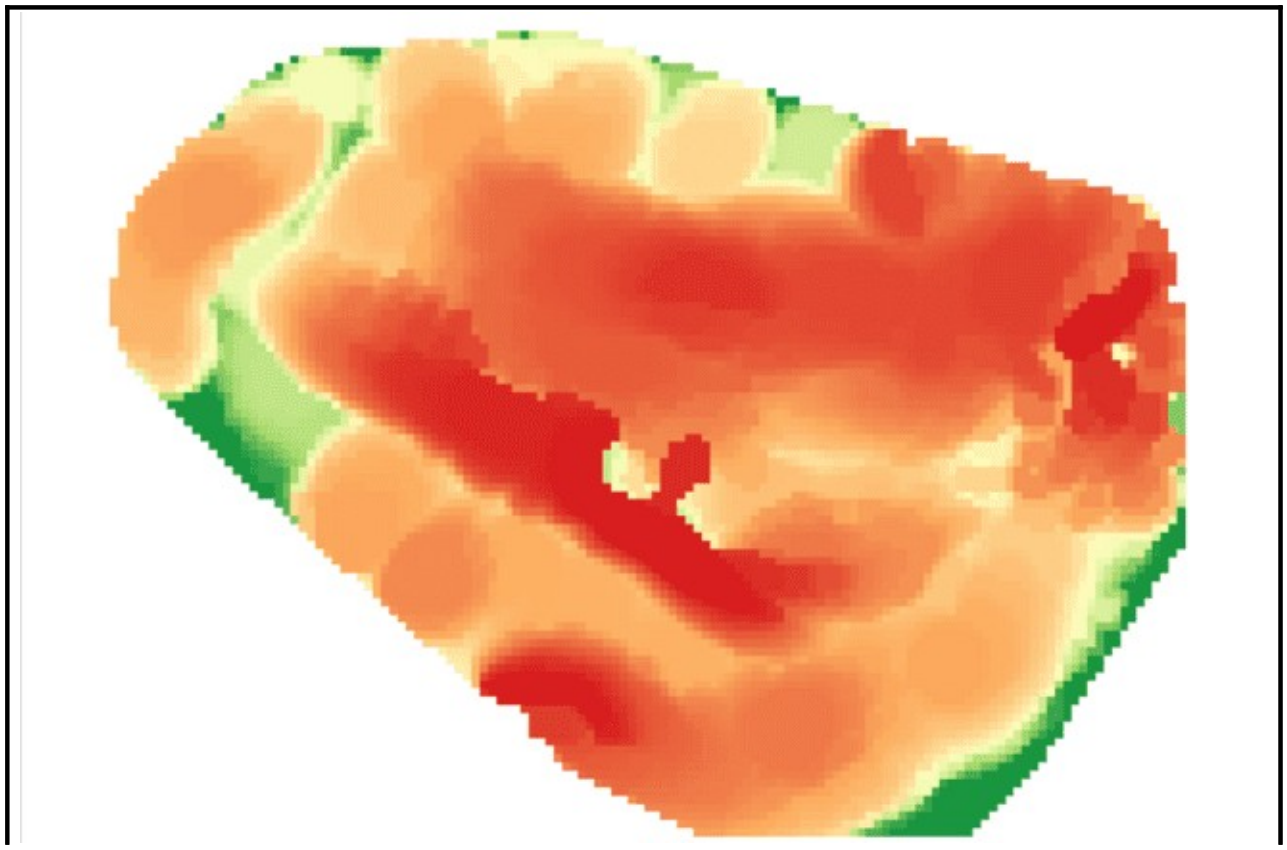


Figure 9: Extreme monsoon saturation (Noamundi Mines setup) simulation heatmap showing orange and red zones across almost the whole pit.

***Geomechanical Discussion*:** The minimum Bishop Factor of Safety is 0.0960, which is far below 1, so the steepest wall sections are unstable in this run. The clay-rich lateritic soils at Noamundi have a low shear envelope (c' = 25.0 kPa, φ' = 24.0°). Under the extreme monsoon saturation coefficient (Ru = 0.40), the soil matrix loses most of its strength and becomes a heavy, unstable mass. The internal friction cannot hold the steep, aggressive Swiss bench cuts anymore, so rotational slumping spreads across almost all active sectors of the pit.

How the Real Mine Layout Copes (From the Photograph)



Figure 10: Photograph of the Noamundi Iron Mine (India), showing a stepped pit with wide benches and haul roads, and regularly spaced white strips running down the foreground bench faces.

The photograph of the Noamundi Mine shows the features below, which help with weak, monsoon-wet material. These are observations from the photograph only, and the angles quoted come from the screening check in Section 6.

- **Flat Overall Slope:** The pit in the photo is stepped, with many benches and wide berms, so its overall slope is much flatter than any single bench face and than the steep Swiss geometry. In the screening check in Section 6, material with c' = 25.0 kPa, φ' = 24.0° and Ru = 0.40 needs about 25° for a single 15 m bench and about 13° for a 150 m wall to keep a FoS of at least 1.3.

- **Wide Benches:** The photo shows wide, flat benches between the bench faces. Wide benches keep the overall angle low and give room to catch material that slumps from the face above, which helps protect the haul roads below.
- **Surface Drainage:** The foreground bench faces have regularly spaced white strips running down the slope, which appear to be lined drainage channels (the photo alone does not confirm this). If they are, they carry runoff off the faces so that water does not soak in or erode them. In the screening check, lowering Ru from 0.40 to 0.10 raises the limits to about 38° for a bench and 19° for the wall.

3.4 Kazakhstan: Bozshakol Mine (Semi-Arid Sedimentary Setup)

***Heatmap Behavioral Pattern*:** Compared with the wet-climate runs, the colours are lighter, with more yellow and orange and less deep red, and patches of green and yellow remain along the left and lower-right margins. Orange and red zones still cover much of the pit interior, with the strongest red along a diagonal band through the pit and at the eastern edge.

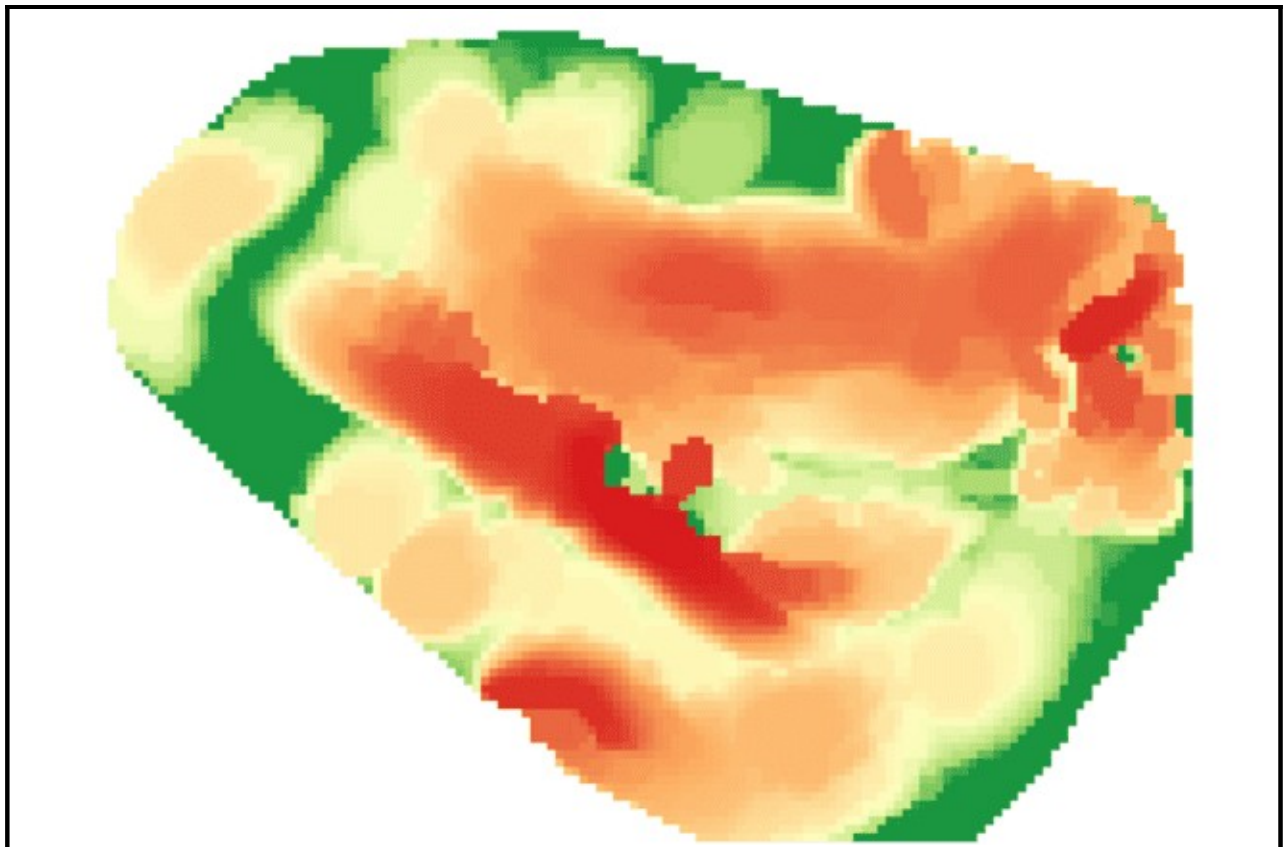


Figure 11: Bozshakol (Kazakhstan) simulation heatmap. Low water pressure makes this run less unstable than the wet-climate runs, but orange and red zones still cover much of the pit.

***Geomechanical Discussion*:** This improvement shows how much dry terrain and lower groundwater (Ru = 0.10) help stability. The sheared sedimentary rock mass has a low cohesion (c' = 25.0 kPa), but its high friction angle (φ' = 34.0°) stays effective because there is very little pore-water pressure to reduce it. With that, this run is clearly less unstable than the wet-climate runs. The minimum Bishop Factor of Safety is still only 0.3632, so the steepest wall sections remain unstable.

How the Real Mine Layout Copes (From the Photograph)



Figure 12: Aerial photograph of the Bozshakol open pit (Kazakhstan), showing an oval pit with curved walls, a ramp and dry surroundings.

The photograph of the Bozshakol pit shows the features below. These are observations from the photograph only, and the angles quoted come from the screening check in Section 6.

- **Curved Pit Walls:** The pit is a large oval and its walls curve around it in plan. In a curved (concave) wall the rock on both sides supports the sliding mass. This is a 3D effect that Scoops3D captures and a 2D section does not, so the 2D numbers in Section 6 are on the safe side for this pit.
- **Wide Benches and a Ramp:** The walls are made of benches with a ramp, so the overall angle is much flatter than a single face. With c' = 25.0 kPa, φ' = 34.0° and Ru = 0.10, the screening check in Section 6 gives about 47° for a single 15 m bench and about 28° for a 150 m wall (FoS of at least 1.3). The strength here comes mainly from friction, because the cohesion is as low as in the India case.
- **Active Slope Management in the Real Pit:** A published study of this pit reports that multi-bench instabilities developed early in mining in a complex, sheared sedimentary rock mass on the

south wall. They were managed with prism and radar monitoring, and 3D back-analysis was used to find realistic shear strength parameters and to support slope optimisation and depressurisation [Adiyansyah et al., 2023].

3.5 Mongolia: Oyu Tolgoi Mine (Arid Seismically Active Setup)

***Heatmap Behavioral Pattern*:** The slopes show high resilience, and soft yellows and safe greens dominate most of the pit. There are, however, narrow high-contrast red lines cutting through the terrain, concentrated at the base of the lowest, steepest bench faces.

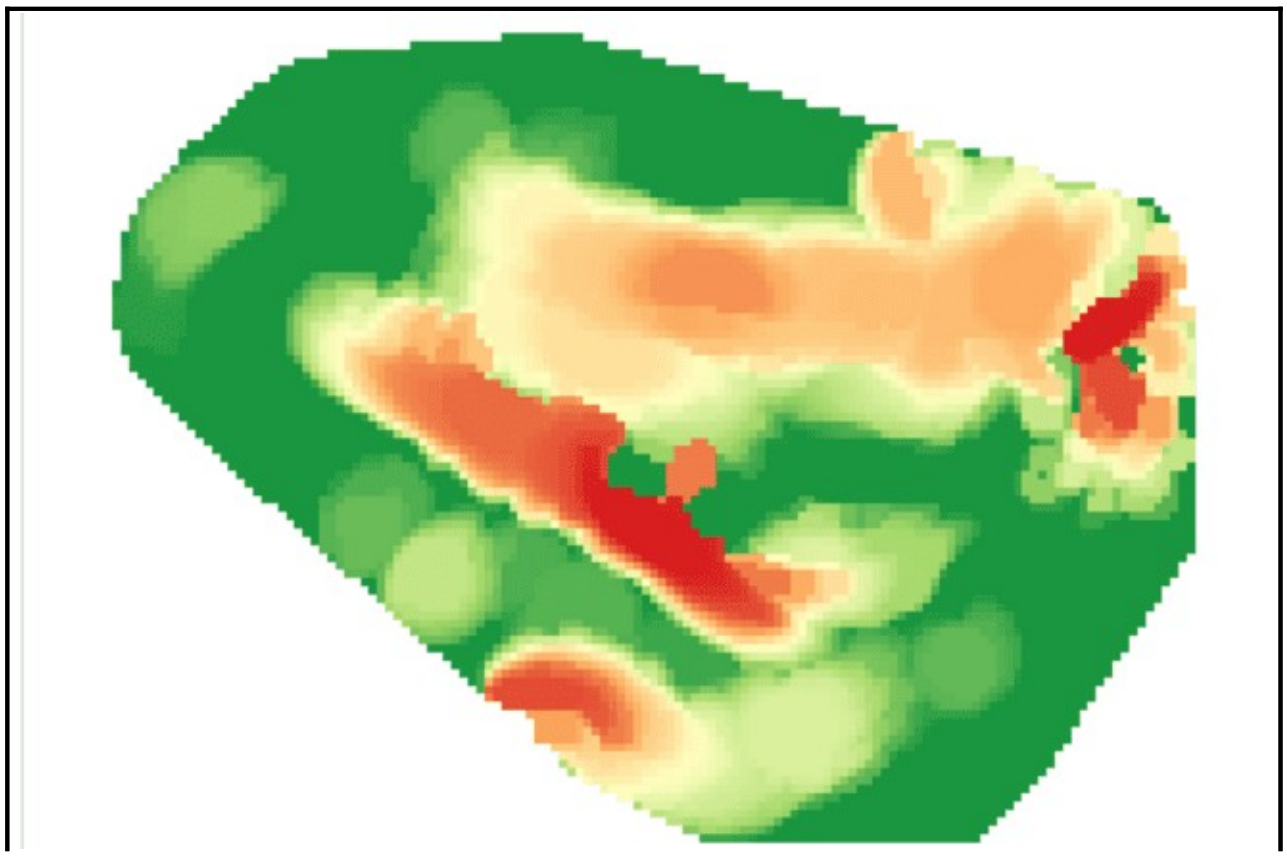


Figure 13: Oyu Tolgoi (Mongolia) simulation heatmap. High rock strength preserves main slope stability, but seismic loading isolates critical risks (red) to the steepest bench bases.

***Geomechanical Discussion*:** With a minimum Bishop Factor of Safety of 1.1441, Mongolia is the only foreign scenario where the steep quarry design technically stays standing (FoS > 1.0). The unweathered Gobi Desert hard-rock mass has very high shear resistance (c' = 180.0 kPa, φ' = 42.0°). Since the model is completely dry (Ru = 0.00), the rock supports the steep slope geometry under normal conditions. The narrow red lines mark the exact spots where the 0.12 horizontal earthquake acceleration concentrates dynamic tectonic stress and causes localized seismic shear failures along the weakest highwall benches.

How the Real Mine Layout Copes (From the Photograph)



Figure 14: Photograph of the Oyu Tolgoi open pit (Mongolia), showing a deep pit with many regular benches, a spiral haul road and a small pond on the pit floor in a dry landscape.

The photograph of the Oyu Tolgoi open pit shows the features below. These are observations from the photograph only, and the angles quoted come from the screening check in Section 6.

- **Many Regular Benches:** The pit walls are made of many regular, closely spaced benches, and the surroundings are dry. In the screening check in Section 6, a 15 m bench in this strong, dry rock ($c' = 180.0$ kPa, $\varphi' = 42.0°$) stays stable even near vertical, and a 150 m wall can be cut at about 56°. With $k_h = 0.12$ the wall limit falls to about 46° (FoS of at least 1.3), so about 10° is given up for the earthquake load.
- **Haul Road and Dry Walls:** A haul road spirals down the wall, and the only water visible is a small pond at the bottom of the pit; the walls themselves look dry, which is consistent with the dry ($R_u = 0.00$) assumption. The photograph does not show rock bolts or mesh on the highwalls, so no ground support is assumed in this study.

- **Underground Block Cave:** Oyu Tolgoi also has an underground block-cave operation (Hugo North), which is separate from the open pit in the photograph and is not modelled in this study [Rwodzi et al., 2026].

4. Quantitative Data Validation Archive

The table below lists the final, verified figures taken directly from the main summary log files of the Scoops3D runs:

```
     LOCATION FOR OUTPUT FILES:
          C:\Users\Sayan Mondal\Documents\pix4d\zambia_nchanga_trial_output\

      zambia_nchanga_trial_out.txt
      zambia_nchanga_trial_errors_out.txt
      zambia_nchanga_trial_slope_out.asc
        Range: [    0.0003,       76.6143]
      zambia_nchanga_trial_fos3d_out.asc
        Range: [    0.1372,      100.0000]
      zambia_nchanga_trial_ordfos3d_out.asc
        Range: [    0.1967,      100.0000]
      zambia_nchanga_trial_fosvol_out.asc
        Range: [  5.0009E+04,  9.9997E+04]
      zambia_nchanga_trial_spheres_out.okc

     Optional files generated:
      zambia_nchanga_trial_filtergrid_out.asc
        Range: [       0,     115]
      zambia_nchanga_trial_filter_out.txt
      zambia_nchanga_trial_ncolerr_out.txt

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
IV. RESULTS:
Number of trial surfaces tried:                                      127450
Number of surfaces eliminated due to filters and/or nonconvergence:     116
     Check file filter_out for detailed information.
Number of surfaces with active column totals less than limcol:       100013
     Check file ncolerr_out for detailed information.
-----------
3D POTENTIAL FAILURE - GLOBAL MINIMUM
Bishop's 3D factor of safety:                                         0.1372
Ordinary 3D factor of safety:                                         0.2063
Volume:                                                          9.42788E+04
Horizontal surface area:                                         3.90166E+03
Slip surface area:                                               8.08427E+03
Weight:                                                          2.16841E+06
Number of active columns:                                                 57
      x-center         y-center         z-center      radius
     311673.3952     5169639.6418         620.0000  1.08471E+02
Slip direction, relative to search lattice:                          55.1218
End date and time: 10/20/2026  10:27:54
Elapsed cpu time in seconds:   1.26250E+01
```

Figure 15: Sample of the output text file generated by Scoops3D. The above output file is named as zambia_nchanga_trial_out.txt

Target Site	Min Bishop FoS	Total Trial Surfaces	Critical Slide Center (Xc, Yc, Zc)	Radius (m)	Volume (V x $10^4 m^3$)	Slip Area (A x $10^3 m^2$)	Processing CPU Time (Seconds)
🇨🇭 **Switzerland**	0.7346	127,450	(311663.3952, 5169669.6418, 630.0000)	129.47	9.849	9.299	9.39
🇿🇲 **Zambia**	0.1372	127,450	(311673.3952, 5169639.6418, 620.0000)	108.5	9.428	8.084	12.62
🇮🇳 **India**	0.0960	127,450	(312123.4, 5169879.6, 590.0)	94.28	5.389	3.918	12.80
🇰🇿 **Kazakhstan**	0.3632	127,450	(311653.3952, 5169669.6418, 620.0000)	115.46	5.094	6.425	12.70
🇲🇳 **Mongolia**	1.1441	127,450	(312113.3952, 5169869.6418, 580.0000)	90.54	6.213	4.352	12.50

Note on active columns (limcol): all five runs used limcol = 100. Scoops3D flagged 76.8% of the trial surfaces in the Zambia run (102,529 of 133,453) and 78.5% in each of the other four runs (100,013 of 127,450) as having fewer active columns than limcol, and listed them in the ncolerr output files. The global-minimum surface of every run is one of these flagged surfaces, and each minimum Bishop FoS in the table above is equal to the lowest F_Bish in the matching ncolerr file. The critical surfaces have only 57 (Zambia), 60 (Switzerland), 38 (India), 43 (Kazakhstan) and 36 (Mongolia) active columns, with horizontal areas of about 2,100 to 4,300 m^2. They are therefore small slides made of a few dozen 10 m DEM cells. The minimum FoS values are the genuine Scoops3D output, but they should be read as indicators of where the steepest wall sections are weak and not as design values. A finer DEM, for example 5 m, would put more columns in each slide.

5. Reconciliation Discussions: Core Design Failures

This comparison shows an important geotechnical point: open-pit slope designs cannot simply be copied from one environment to another.

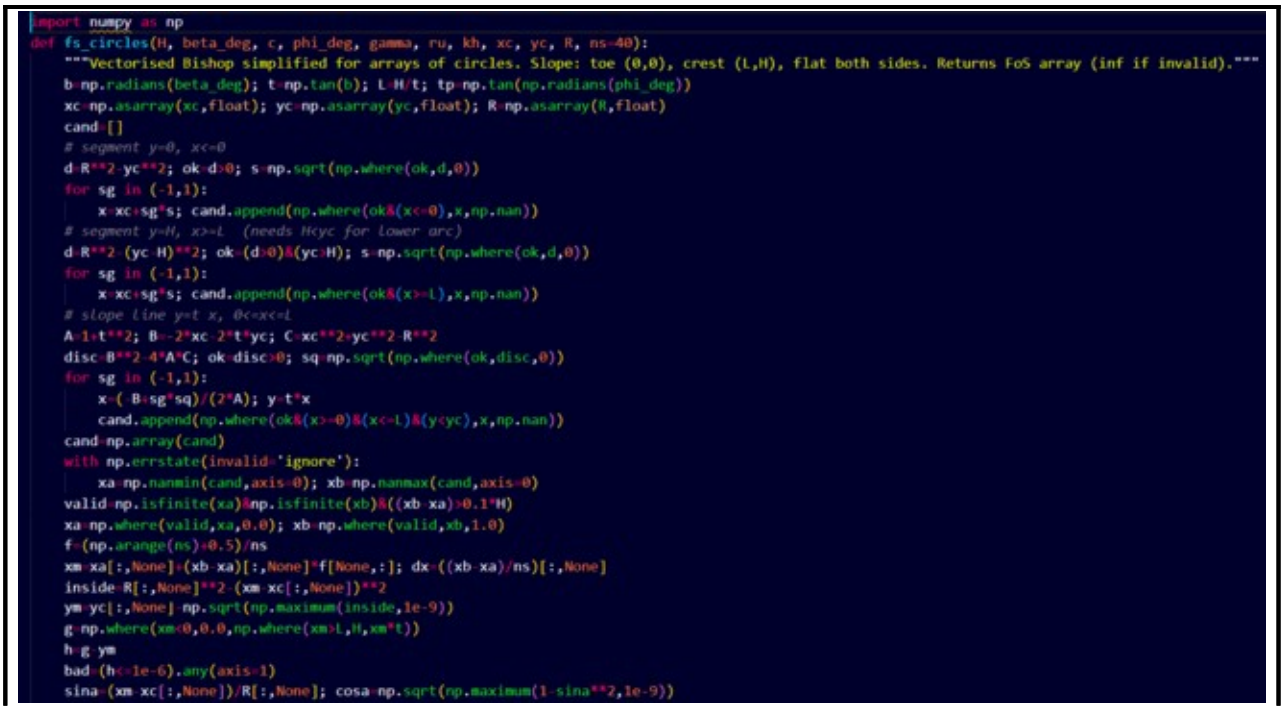
The Swiss Eclépens Quarry layout has a tight, aggressive bench design with steep slope angles, which suits massive, high-cohesion limestone with a low groundwater risk. Even so, the Swiss baseline itself reaches a minimum Bishop Factor of Safety of 0.7346 on its steepest walls (Section 3.1). When I apply this same geometry to the other environments without changing it, each one is affected by a specific geological or climatic factor:

***1.The Fluid-Dynamic Failure (Zambia & India)*:** In tropical and monsoon climates the steep Swiss bench cuts cannot be copied. High groundwater pressures (Ru ≥ 0.35) greatly weaken low-cohesion soils and weathered shales, cut their effective shear strength sharply, and trigger rotational failures on the

steepest walls (minimum Bishop FoS of 0.0960 for India and 0.1372 for Zambia).
2. *The Low-Cohesion Structural Block (Kazakhstan)*: Low water levels help, but the minimum Bishop Factor of Safety is still 0.3632 because the weak cohesion (c' = 25.0 kPa) of sheared sedimentary rock cannot hold steep, high-angle bench cuts for long, and this risks rapid progressive multi-bench failures along natural structural joints.
3. *The Dynamic Tectonic Load (Mongolia):* Ultra-strong Gobi rock easily holds the steep Swiss layout under dry, static conditions, but the earthquake load cuts the safety margin sharply. The horizontal earthquake acceleration factor (kh = 0.12) puts large inertial forces on the steep bench profiles, so the minimum Bishop Factor of Safety is only 1.1441 and localized seismic shear failures concentrate on the weakest highwall benches.

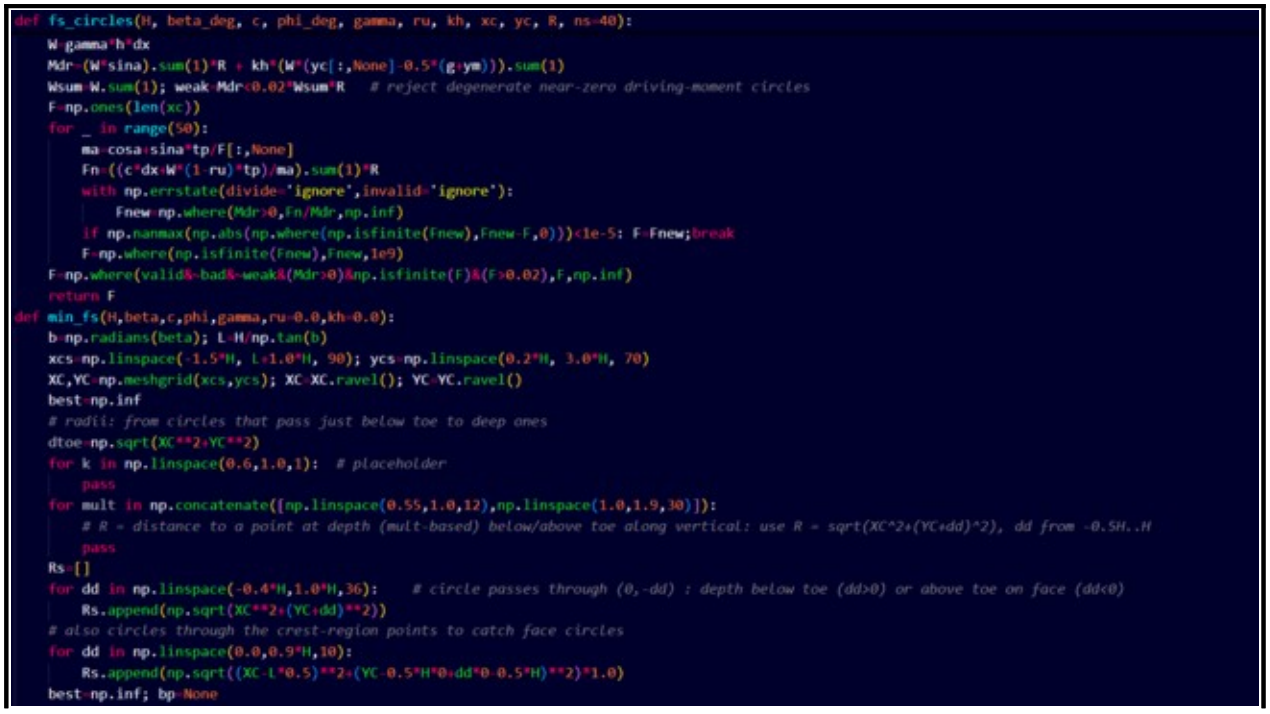
6. Indicative Slope-Angle Check on the Real Mine Layouts

The photographs in Section 3 show how the real mines are laid out, but a photograph cannot give bench widths or slope angles. To put numbers on the effect of the layout, a simple 2D Bishop's Simplified check [Bishop, 1955] (uniform slope, circular failure) was run with the c', φ', γ, Ru and kh values from Section 2. For each environment I found the steepest slope angle that still gives a Factor of Safety of at least 1.3, for a single 15 m bench and for a 150 m overall wall. Both heights are assumed values. The check reproduces Taylor's stability numbers [Taylor, 1948] for a φ = 0 slope (0.191 at 60° and 0.181 at 45°) and the infinite-slope result for a cohesionless slope. It is only a screening check and it does not replace the Scoops3D results, because it ignores 3D effects, layering and joints.



```
import numpy as np
def fs_circles(H, beta_deg, c, phi_deg, gamma, ru, kh, xc, yc, R, ns=40):
    """Vectorised Bishop simplified for arrays of circles. Slope: toe (0,0), crest (L,H), flat both sides. Returns FoS array (inf if invalid)."""
    b=np.radians(beta_deg); t=np.tan(b); L=H/t; tp=np.tan(np.radians(phi_deg))
    xc=np.asarray(xc,float); yc=np.asarray(yc,float); R=np.asarray(R,float)
    cand=[]
    # segment y=0, x<=0
    d=R**2-yc**2; ok=d>0; s=np.sqrt(np.where(ok,d,0))
    for sg in (-1,1):
        x=xc+sg*s; cand.append(np.where(ok&(x<=0),x,np.nan))
    # segment y=H, x>=L  (needs Hcyc for lower arc)
    d=R**2-(yc-H)**2; ok=(d>0)&(yc>H); s=np.sqrt(np.where(ok,d,0))
    for sg in (-1,1):
        x=xc+sg*s; cand.append(np.where(ok&(x>=L),x,np.nan))
    # slope line y=t*x, 0<=x<=L
    A=1+t**2; B=-2*xc-2*t*yc; C=xc**2+yc**2-R**2
    disc=B**2-4*A*C; ok=disc>0; sq=np.sqrt(np.where(ok,disc,0))
    for sg in (-1,1):
        x=(-B+sg*sq)/(2*A); y=t*x
        cand.append(np.where(ok&(x>=0)&(x<=L)&(y<yc),x,np.nan))
    cand=np.array(cand)
    with np.errstate(invalid='ignore'):
        xa=np.nanmin(cand,axis=0); xb=np.nanmax(cand,axis=0)
    valid=np.isfinite(xa)&np.isfinite(xb)&((xb-xa)>0.1*H)
    xa=np.where(valid,xa,0.0); xb=np.where(valid,xb,1.0)
    f=(np.arange(ns)+0.5)/ns
    xm=xa[:,None]+(xb-xa)[:,None]*f[None,:]; dx=((xb-xa)/ns)[:,None]
    inside=R[:,None]**2-(xm-xc[:,None])**2
    ym=yc[:,None]+np.sqrt(np.maximum(inside,1e-9))
    g=np.where(xm<0,0.0,np.where(xm>L,H,xm*t))
    h=g-ym
    bad=(h<=1e-6).any(axis=1)
    sina=(xm-xc[:,None])/R[:,None]; cosa=np.sqrt(np.maximum(1-sina**2,1e-9))
```

Figure 16: Vectorized geometry intersection and slice discretization logic for the simplified Bishop method (fs_circles).



```
def fs_circles(H, beta_deg, c, phi_deg, gamma, ru, kh, xc, yc, R, ns=40):
    W=gamma*h*dx
    Mdr=(W*sina).sum(1)*R + kh*(W*(yc[:,None]-0.5*(g+ym))).sum(1)
    Wsum=W.sum(1); weak=Mdr<0.02*Wsum*R   # reject degenerate near-zero driving-moment circles
    F=np.ones(len(xc))
    for _ in range(50):
        ma=cosa+sina*tp/F[:,None]
        Fn=((c*dx+W*(1-ru)*tp)/ma).sum(1)*R
        with np.errstate(divide='ignore',invalid='ignore'):
            Fnew=np.where(Mdr>0,Fn/Mdr,np.inf)
        if np.nanmax(np.abs(np.where(np.isfinite(Fnew),Fnew-F,0)))<1e-5: F=Fnew;break
        F=np.where(np.isfinite(Fnew),Fnew,1e9)
    F=np.where(valid&~bad&~weak&(Mdr>0)&np.isfinite(F)&(F>0.02),F,np.inf)
    return F
def min_fs(H,beta,c,phi,gamma,ru=0.0,kh=0.0):
    b=np.radians(beta); L=H/np.tan(b)
    xcs=np.linspace(-1.5*H, L+1.0*H, 90); ycs=np.linspace(0.2*H, 3.0*H, 70)
    XC,YC=np.meshgrid(xcs,ycs); XC=XC.ravel(); YC=YC.ravel()
    best=np.inf
    # radii: from circles that pass just below toe to deep ones
    dtoe=np.sqrt(XC**2+YC**2)
    for k in np.linspace(0.6,1.0,1):  # placeholder
        pass
    for mult in np.concatenate([np.linspace(0.55,1.0,12),np.linspace(1.0,1.9,30)]):
        # R = distance to a point at depth (mult-based) below/above toe along vertical: use R = sqrt(XC^2+(YC+dd)^2), dd from -0.5H..H
        pass
    Rs=[]
    for dd in np.linspace(-0.4*H,1.0*H,36):    # circle passes through (0,-dd) : depth below toe (dd>0) or above toe on face (dd<0)
        Rs.append(np.sqrt(XC**2+(YC+dd)**2))
    # also circles through the crest-region points to catch face circles
    for dd in np.linspace(0.0,0.9*H,10):
        Rs.append(np.sqrt((XC-L*0.5)**2+(YC-0.5*H*0+dd*0-0.5*H)**2)*1.0)
    best=np.inf; bp=None
```

Figure 17: Factor of Safety fixed-point iteration loop and candidate trial-circle generation for global slope search.

```
def min_fs(H,beta,c,phi,gamma,ru=0.0,kh=0.0):
    for R in Rs:
        F=fs_circles(H,beta,c,phi,gamma,ru,kh,XC,YC,R)
        i=np.argmin(F)
        if F[i]<best: best=F[i]; bp=(XC[i],YC[i],R[i])
    # Local refinement (Nelder-Mead)
    from scipy.optimize import minimize
    def obj(p):
        v=fs_circles(H,beta,c,phi,gamma,ru,kh,[p[0]],[p[1]],[p[2]])[0]
        return min(v,50.0)
    if bp is not None:
        r=minimize(obj,bp,method='Nelder-Mead',options={'xatol':0.05,'fatol':1e-4,'maxiter':400})
        best=min(best,r.fun)
    return best
if __name__=="__main__":
    import time; t0=time.time()
    print('cohesionless dry phi30 b26.57:',round(min_fs(50,26.57,0,30,20,0),3),'(expect ~1.156)')
    print('ru=0.2 same:',round(min_fs(50,26.57,0,30,20,0.2),3),'(infinite-slope 0.867)')
    print('phi=0 c/gH=0.1 b60 (Taylor Ns=0.191 -> 0.524):',round(min_fs(50,60,100,0,20,0),3))
    print('phi=0 c/gH=0.1 b45 (Taylor Ns=0.170 -> 0.588):',round(min_fs(50,45,100,0,20,0),3))
    print('Bishop-Morgenstern c/gH=.05 phi=30 cotb=2 ru=0 (chart m~1.9):',round(min_fs(50,26.57,50,30,20,0),3))
    print('time',round(time.time()-t0,1))
```

Figure 18: Minimum Factor of Safety finder using Nelder-Mead optimization (scipy.optimize) and benchmark validation test suite.

```
c:\Users\Sayan Mondal\Downloads\slope_check.py:23: RuntimeWarning: All-NaN slice encountered
  xa=np.nanmin(cand,axis=0); xb=np.nanmax(cand,axis=0)
cohesionless dry phi30 b26.57: 1.155 (expect ~1.156)
ru=0.2 same: 0.866 (infinite-slope 0.867)
phi=0 c/gH=0.1 b60 (Taylor Ns=0.191 -> 0.524): 0.524
phi=0 c/gH=0.1 b45 (Taylor Ns=0.170 -> 0.588): 0.551
Bishop-Morgenstern c/gH=.05 phi=30 cotb=2 ru=0 (chart m~1.9): 1.885
time 29.9
```

Figure 19: Terminal execution output showing benchmark validation comparisons against analytical and chart-based Factor of Safety solutions.

6.1 Limiting Slope Angles

```python
import sys, warnings; warnings.filterwarnings('ignore'); sys.path.insert(0,'.')
from slope_check import min_fs
from concurrent.futures import ProcessPoolExecutor
cases = {
 'Switzerland':(140,38,25.5,0.20,0.0),
 'Zambia':(35,27,23.0,0.35,0.0),
 'India':(25,24,21.5,0.40,0.0),
 'Kazakhstan':(25,34,24.5,0.10,0.0),
 'Mongolia static':(180,42,26.5,0.0,0.0),
 'Mongolia kh=0.12':(180,42,26.5,0.0,0.12),
 'Zambia drained Ru=0.10':(35,27,23.0,0.10,0.0),
 'India drained Ru=0.10':(25,24,21.5,0.10,0.0),
}
def maxang(a):
    name,H,target=a
    c,phi,g,ru,kh=cases[name]
    lo,hi=8.0,80.0
    if min_fs(H,lo,c,phi,g,ru,kh)<target: return (name,H,target,'<8')
    if min_fs(H,hi,c,phi,g,ru,kh)>=target: return (name,H,target,'>80')
    for _ in range(7):
        mid=(lo+hi)/2
        if min_fs(H,mid,c,phi,g,ru,kh)>=target: lo=mid
        else: hi=mid
    return (name,H,target,round(lo,1))
jobs=[(n,H,1.3) for n in cases for H in (15,150)]
jobs+=[('Mongolia kh=0.12',15,1.1),('Mongolia kh=0.12',150,1.1)]
if __name__=="__main__":
    with ProcessPoolExecutor() as ex:
        for r in ex.map(maxang,jobs): print(r,flush=True)
```

Figure 20: Multi-country slope stability analysis runner (run_table.py) using binary search for critical slope angle determination with parallel processing via ProcessPoolExecutor

Site	Strength / water condition	15 m bench (FoS ≥ 1.3)	150 m wall (FoS ≥ 1.3)	With drainage or seismic load
Switzerland (Baseline)	c' = 140.0 kPa, φ' = 38.0°, Ru = 0.20	>80°	36°	-
India (Noamundi)	c' = 25.0 kPa, φ' = 24.0°, Ru = 0.40	25°	13°	Ru = 0.10: 38° bench, 19° wall
Zambia (Nchanga)	c' = 35.0 kPa, φ' = 27.0°, Ru = 0.35	36°	16°	Ru = 0.10: 49° bench, 23° wall
Kazakhstan (Bozshakol)	c' = 25.0 kPa, φ' = 34.0°, Ru = 0.10	47°	28°	Curved walls add 3D support (not counted here)
Mongolia (Oyu Tolgoi)	c' = 180.0 kPa, φ' = 42.0°, Ru = 0.00, kh = 0.12	>80°	56° static, 46° with kh = 0.12	kh = 0.12 with FoS ≥ 1.1: 56°

Note: angles are rounded to the nearest degree. ">80°" means the slope was still stable at the steepest angle tested (80°). The values are screening results for an assumed uniform rock mass and are not the design angles of the real mines.

6.2 What the Numbers Show

India: At Ru = 0.40 the weak lateritic material allows only about 25° for a single bench and 13° for a 150 m wall at a FoS of 1.3, which is why Noamundi needs a flat overall angle and wide benches. Drainage matters as much as geometry: bringing Ru down from 0.40 to 0.10 raises the limits to about 38° and 19°, a gain of about 13° for a bench and 6° for the wall without changing the soil strength.

Zambia: Nchanga gives about 36° and 16°, and drainage to Ru = 0.10 raises them to about 49° and 23°. The wide benches in the photograph and the drainage of water from the walls act on exactly these two limits.

Kazakhstan: With φ' = 34.0° and Ru = 0.10 the limits are about 47° and 28°. The cohesion is as low as in India, but the limits are much higher, so the strength here comes mainly from friction. The 2D check ignores the curved plan shape of a pit like this, so its numbers are on the safe side.

Mongolia: A 15 m bench stays stable even near vertical, and a 150 m wall can be cut at about 56°. Applying kh = 0.12 lowers the FoS = 1.3 limit to about 46°, so roughly 10° is given up for the earthquake load. If a lower pseudo-static FoS of 1.1 is accepted, the limit returns to about 56°.

Taken together, wide benches that keep the overall angle low and drainage that lowers Ru are the two layout features that change the limits the most in the weak environments, while in the strong Gobi rock the earthquake load costs only about 10°.

6.3 2D Limits and 3D Minimum Bishop FoS Values

The two checks answer different questions. The Scoops3D runs apply the unchanged Swiss quarry DEM, whose steepest faces reach about 77°, to each environment's weak strength values. That is why the minimum Bishop FoS falls far below 1 for Zambia (0.1372) and India (0.0960). The 2D check instead asks what slope angle each environment can actually hold at a FoS of 1.3, and for the weak, wet cases the answer is only about 13° to 36°. A 77° face is far steeper than any of those limits, so a very low 3D FoS and a "realistic" 2D design angle are consistent with each other. The 3D minimum also comes from a small slide on one steep wall section (note on active columns in Section 4), not from the pit as a whole.

7. Academic Bibliography & References

Part A: Photographic & Media Source Link References

- **Figure 8 Source (Zambia - Nchanga Mine):**
  Financial Times. (2019). *Konkola Copper Mines operations and regional economic impact*. Financial Times Commodities Archive. ft.com

- **Figure 10 Source (India - Noamundi Mine):**
  Tata Steel Limited. (2025). *Noamundi@100: A century of strength, sustainability, and stewardship*. Tata Steel Integrated Report and Annual Accounts 2024-25. tatasteel.com
- **Figure 12 Source (Kazakhstan - Bozshakol Mine):**
  KAZ Minerals. (2026). *Bozshakol: A low cost copper mine in Pavlodar, Kazakhstan*. Our Locations: Operations Profile. kazminerals.com
- **Figure 14 Source (Mongolia - Oyu Tolgoi Mine):**
  Weidman, T. (2018, September 22). *An open pit mine is seen at the Oyu Tolgoi copper-gold mine* [Editorial News Photograph]. Bloomberg via Getty Images.

Part B: Primary Academic & Geotechnical Literature

- **Kazakhstan (Bozshakol Mine):**
  Adiyansyah, B., Gorokhov, D., Rakhmetov, B., Caratti, J., & Bar, N. (2023). Three-dimensional geotechnical analysis on a strongly altered sedimentary domain at Bozshakol copper mine. In P. M. Dight (Ed.), *SSIM 2023: Third International Slope Stability in Mining Conference* (pp. 965–982). Australian Centre for Geomechanics. https://doi.org/10.36487/ACG_repo/2335_66
- **Mongolia (Oyu Tolgoi Block Cave):**
  Rwodzi, L., Ekkerd, J., Taylor, M., Batsaikhan, A., Russo, A., Ramírez, C., Hormazábal, E., & Nur, J. (2026). Geotechnical characterisation of Hugo North Lift 1, Oyu Tolgoi. In *Caving 2026: Proceedings of the Sixth International Conference on Block and Sublevel Caving* (pp. 1–19). Australian Centre for Geomechanics.
- **Methodological Framework (The Software Engine):**
  Reid, M. E., Christian, S. B., Brien, D. L., & Henderson, S. T. (2015). *Scoops3D—Software to analyze 3D slope stability throughout a digital landscape* (Techniques and Methods, book 14, chap. A1, 218 p.). U.S. Geological Survey. https://doi.org/10.3133/tm14A1
- **Methodological Framework (2D Slope Check):**
  Bishop, A. W. (1955). The use of the slip circle in the stability analysis of slopes. *Géotechnique*, 5(1), 7–17.
  Taylor, D. W. (1948). *Fundamentals of Soil Mechanics*. John Wiley & Sons.

Part C: Software Tools & AI Assistance

- **Photogrammetry, GIS and Slope-Stability Software:**
  - Pix4D SA. *Pix4Dmapper* [Photogrammetry software]. https://www.pix4d.com
  - QGIS Development Team. *QGIS Geographic Information System* [Software]. https://qgis.org

    - U.S. Geological Survey. *Scoops3D (version 1.3.01)* [Software]. https://www.usgs.gov/software/scoops3d
    - Harris, C. R., et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362.
    - Virtanen, P., et al. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272.
- **Data Synthesis & Parameter Benchmarking:**
    - Pix4D SA. (n.d.). *Example projects – real photogrammetry data* (Quarry dataset) [Dataset]. Pix4D Support. https://support.pix4d.com/hc/en-us/articles/360000235126
    - Google. (2026). *Google AI Mode* [Synthesized geotechnical parameter matrix for comparative slope stability analysis]. Retrieved from https://www.google.com
- **Code Generation:**
    - Anthropic. (2026). *Claude* [Large language model used for coding and script generation]. Retrieved from https://claude.ai